HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 03 Name: _____ Marks: / 34

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The scale of a map is $1 : 50000$. If the actual area of an airport is $10\,km^2$, then the area of this airport on the map is

   A. $2\,cm^2$.

   B. $4\,cm^2$.

   C. $20\,cm^2$.

   D. $40\,cm^2$.

(DSE-2023 PII Q12)

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2. Let a_n be the nth term of a sequence. If $a_6 = 23$, $a_8 = 60$ and $a_{n+2} = a_{n+1} + a_n$ for any positive integer n, then $a_3 =$

A. 4.

B. 5.

C. 9.

D. 14.

(DSE-2023 PII Q14)

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3. The length of a side of a solid cube is $60\,\text{cm}$. The volume of a solid right circular cylinder is equal to the volume of the cube while the curved surface area of the circular cylinder is equal to the total surface area of the cube. Find the base radius of the circular cylinder.

A. $20\,\text{cm}$

B. $30\,\text{cm}$

C. $76\,\text{cm}$

D. $172\,\text{cm}$

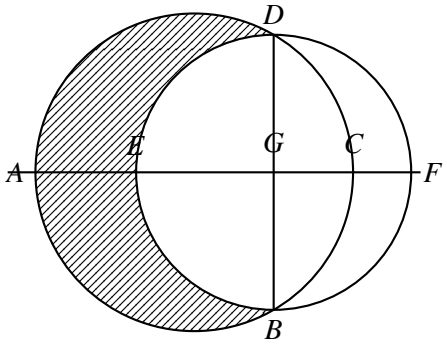
(DSE-2023 PII Q15)

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4. In the figure, AC is a diameter of the circle $ABCD$ while BD and EF are diameters of the circle $BEDF$. It is given that C and E lie on AF. Let G be the point of intersection of AF and BD. If $AG = 30\,\text{cm}$ and $CG = 10\,\text{cm}$, find the area of the shaded region correct to the nearest cm^2.



A. $209\,\text{cm}^2$

B. $367\,\text{cm}^2$

C. $383\,\text{cm}^2$

D. $540\,\text{cm}^2$

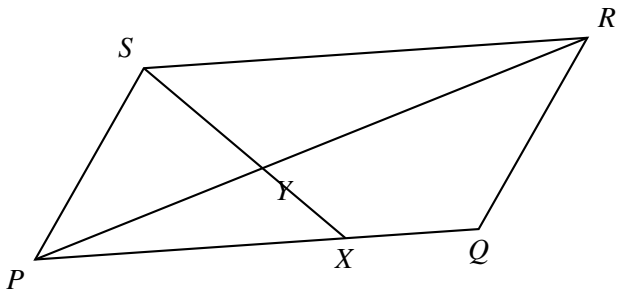
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5. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,cm^2$ and $58\,cm^2$ respectively, then the area of $\triangle RSY$ is



A. $40\,cm^2$.

B. $50\,cm^2$.

C. $58\,cm^2$.

D. $72\,cm^2$.

(DSE-2023 PII Q17)

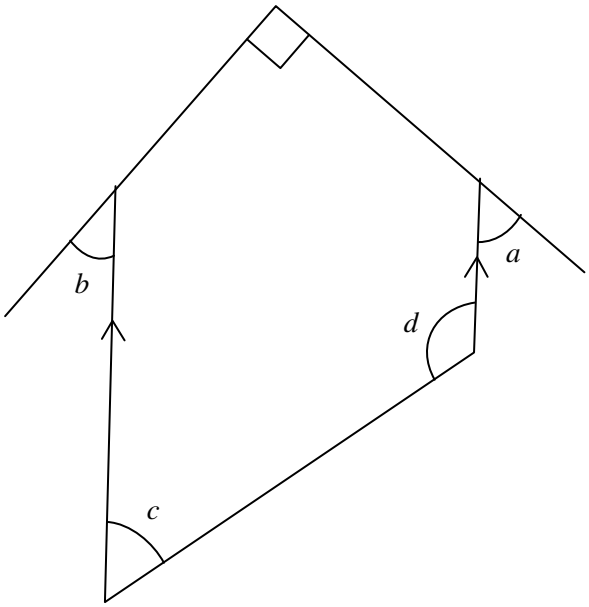
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6. According to the figure, which of the following must be true?

?



I. $a+b=90^\circ$

II. $c+d=180^\circ$

III. $a+b+c=d$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

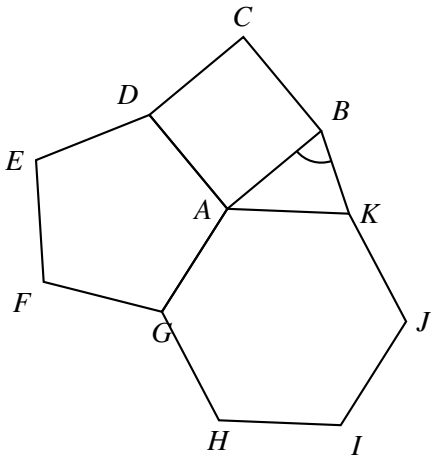
(DSE-2023 PII Q18)

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7. The figure shows the square $ABCD$, the regular pentagon $ADEFG$ and the regular hexagon $AGHIJK$. Find $\angle ABK$.



A. 69°

B. 72°

C. 74°

D. 75°

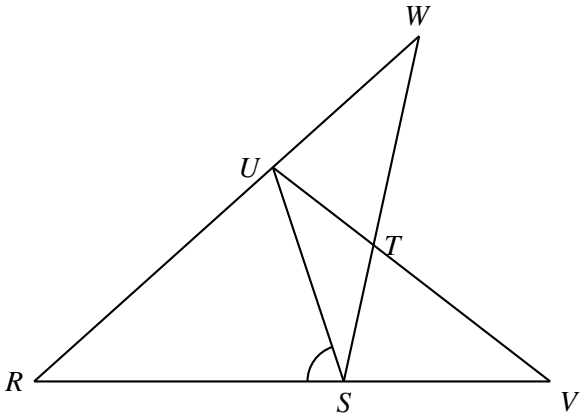
(DSE-2023 PII Q20)

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8. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. RS produced and UT produced meet at the point V while RU produced and ST produced meet at the point W. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$



A. 65°.

B. 73°.

C. 80°.

D. 82°.

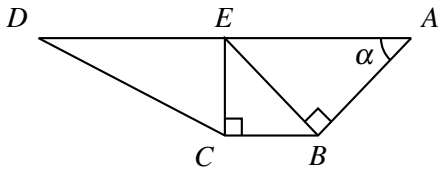
(DSE-2023 PII Q22)

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9. In the figure, $ABCD$ is a trapezium with $AD//BC$. Let E be the mid-point of AD. It is given that $\angle ABE = \angle BCE = 90°$. Find $\dfrac{CE}{DE}$.



A. $\dfrac{1}{2}$

B. 1

C. $\tan\alpha$

D. $\sin\alpha\cos\alpha$

(DSE-2023 PII Q23)

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10. It is given that $\log_8 y$ is a linear function of $\log_4 x$. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 5 and 3 respectively. Which of the following must be true?

A. $x^5y^2 = 8^{10}$

B. $x^6y^5 = 8^{20}$

C. $x^{10}y^3 = 8^{20}$

D. $x^9y^{10} = 8^{30}$

(DSE-2023 PII Q33)

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11. Let $f(x) = 3x^2 + 18mx + 22m^2$, where m is a real constant. Which of the following statements about the graph of $y = -f(3x)$ must be true?

I. The x-coordinate of the vertex of the graph is m.

II. The y-coordinate of the vertex of the graph is $5m^2$.

III. The equation of the axis of symmetry of the graph is $x + m = 0$.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q35)

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12. Let $T(n)$ be the nth term of an arithmetic sequence. If $T(11) = 83$ and $T(25) + T(30) = 463$, find the least value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) > 4 \times 10^5$.

A. 299

B. 300

C. 944

D. 945

(DSE-2023 PII Q36)

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13. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

(DSE-2023 PII Q37)

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14. The straight line $hx + ky = 6$ and the circle $x^2 + y^2 - 8x - 4y - 18 = 0$ intersect at the points M and N, where h and k are constants. If the coordinates of the mid-point of MN are $(1, 0)$, find k.

A. 4

B. 6

C. 9

D. 12

(DSE-2023 PII Q39)

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15. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. 0.

(DSE-2023 PII Q40)

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16. The table below shows the scores (in marks) and the corresponding standard scores of three students in an examination.

Score (marks)	46	x	86
Standard score	-3	1	2

Find x.

A. 64

B. 66

C. 70

D. 78

(DSE-2023 PII Q44)

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17. It is given that n is an integer. Let u, v and w be the standard deviation, the median and the range of the group of numbers $\{1-9n, 3-9n, 4-9n, 5-9n, 7-9n\}$ respectively. Which of the following must be true?

I. $u = 2$

II. $v < 4$

III. $w > 6$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q45)

I can finish _____ questions

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